Joint Spatio-Temporal Subgraph Filter Learning

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***Abstract*—In numerous graph signal processing applications, the availability of data is often restricted to a subset of vertices, necessitating the development of methods for processing and analyzing signals on subgraphs. Subgraph filter learning offers a systematic framework for deriving filters that operate on subgraphs while maintaining their correspondence to the ambient graph. However, existing subgraph filter banks predominantly focus on spatial relationships and overlook temporal dependencies inherent in sequential graph signals. In this work, we introduce a joint vertex–time tensor filter bank that seamlessly integrates spatial information propagation across nodes with causal temporal dependencies across time snapshots. This approach yields a structured and interpretable operator space, enabling more effective spatio-temporal signal processing. Experimental evaluations on both synthetic datasets and real-world traffic flow data demonstrate that the proposed joint-domain framework consistently outperforms spatial-only subgraph filter learning methods, achieving superior performance under comparable parameter constraints.**

***Index Terms*—graph signal processing, time series, vertex-time domain, subgraph filter learning**

I. Introduction

Graph signal processing (GSP) has emerged as a versatile framework for analyzing structured data defined on irregular domains, encompassing a broad range of topics such as graph neural networks [1], [2], graph spectral analysis [3], and topology inference [4], [5]. The graph shift operator (GSO) is a crucial concept which encodes the underlying graph topology and serves as the foundation for many fundamental operations. In particular, access to an appropriate GSO is essential for tasks including graph filtering [6], [7], signal recovery [8], and the computation of the graph Fourier transform (GFT) [9].

Most classical signal processing tasks in GSP, such as filtering and spectral analysis, are developed under the assumption that the complete ambient graph structure is available. In practice, this assumption is often violated, and one may only have access to a partial graph signal $\mathbf{x}_0$ supported on a subset of nodes $V_{\mathrm{sub}} \subset V$. Directly restricting graph filters or spectral tools to the induced subgraph generally leads to structural and spectral mismatches, resulting in information loss. To address this challenge, subgraph filter learning (SFL) has been proposed as an effective framework for processing subgraph signals [10]. The goal of SFL is to learn filters supported on V_{sub} that approximate the action of an ambient graph filter restricted to the same subset, without requiring access to the full ambient graph.

Despite its effectiveness in handling partial observability, the established SFL framework is typically formulated under the assumption that input graph signals are independent samples drawn from a fixed data distribution. Consequently, temporal relationships among successive graph signals are not explicitly modeled. In many application domains, including transaction networks, social media interactions, sensor networks, and traffic systems, graph signals exhibit explicit or implicit temporal dependencies. Omitting such temporal structure when designing subgraph filters may result in a mismatch between the learned filter and the evolving data distribution, yielding unsatisfactory performance. The following example illustrates this limitation.

Example 1. *Consider a stakeholder managing a transaction network that evolves over time and employs a proprietary graph filter for downstream analytical tasks. Due to privacy constraints, a third party is granted access only to a subset of nodes (clients) and aims to replicate the filtering process using this partial data. The goal is to ensure that the filtered outputs on the accessible nodes closely approximate those obtained by applying the full graph filter to the complete data, followed by restricting the result to the same subset.*

Conventional SFL methodologies typically learn static subgraph filters without accounting for temporal dependencies. However, in transaction networks, the distribution of graph signals often exhibits temporal dynamics, leading to potential distribution shifts over time. As a result, filters learned from aggregated historical snapshots may fail to generalize effectively to future data. This limitation underscores the necessity for a more expressive subgraph filter framework that explicitly incorporates temporal structure to enhance adaptability and performance in dynamic environments.

This paper presents a novel formulation of the SFL problem, termed joint spatio-temporal subgraph filter learning (JST-SFL), which extends subgraph filter learning to the joint vertex-time domain. Leveraging access to the ambient graph filter and full spatio-temporal graph signals during training, the proposed framework aims to learn a family of subgraph filters defined on a specified subset of nodes. These filters are designed to optimize performance along the temporal dimension by incorporating causal temporal filtering mechanisms. By explicitly modeling temporal dependencies, the proposed approach effectively addresses temporal distribution shifts, enhancing adaptability in dynamic environments. Importantly, the learned subgraph filters preserve the privacy of the ambient graph filter and the unobserved graph structure while offering improved flexibility and approximation capabilities in spatio-temporal signal processing tasks.

II. Preliminaries

A. Graph Signal Processing

Let $G = (V, E)$ be an unweighted graph, where V denotes the set of vertices and E represents the set of edges. A spatio-temporal graph signal $\{\mathbf{x}_t\}_{t\in\mathbb{N}}$ on G is defined as a sequence of graph signals indexed by discrete time t. For a given time interval $[t_1, t_2)$, where $t_1, t_2 \in \mathbb{N}$ and $t_1 < t_2$, the signal fragment $\mathbf{x}_{t_1:t_2}$ is represented as a matrix in $\mathbb{R}^{|V|\times(t_2-t_1)}$, with each column corresponding to the graph signal at a specific time step within the interval.

The Laplacian matrix of G, denoted by $\mathbf{L} \in \mathbb{R}^{|V|\times|V|}$, is given by $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D} = \operatorname{diag}\{\deg(v_i)\}_{i=1}^{|V|}$ is the degree matrix and $\mathbf{A}$ is the adjacency matrix. The symmetric normalized Laplacian is defined as $\widetilde{\mathbf{L}} = (\mathbf{D}^\dagger)^{1/2}\mathbf{L}(\mathbf{D}^\dagger)^{1/2}$, where $\mathbf{D}^\dagger$ denotes the Moore-Penrose inverse of $\mathbf{D}$. Since $\mathbf{L}$ is symmetric and positive semi-definite, it admits an orthonormal eigendecomposition $\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\intercal$, which allows defining generalized powers $\mathbf{L}^p = \mathbf{U}\mathbf{\Lambda}^p\mathbf{U}^\intercal$ for any $p \in \mathbb{R}_+$; $\widetilde{\mathbf{L}}^p$ is defined analogously.

For any pair of vertices $u, v \in V$, let $d_G(u, v)$ denote the shortest-path distance between u and v in G. The diameter of G is then defined as $\operatorname{diam}(G) = \max\{d_G(u, v) : u, v \in V\}$.

Given a subset $V_{\text{sub}} \subset V$, the induced subgraph is defined as $G_{\text{sub}} \coloneqq (V_{\text{sub}}, \{(u, v) \in E : u, v \in V_{\text{sub}}\})$. Throughout this paper, all subgraphs are assumed to be induced subgraphs unless explicitly stated otherwise. For a spatio-temporal graph signal $\{\mathbf{x}_t\}_{t\in\mathbb{N}}$ defined on G, its restriction to V_{sub} is referred to as a spatio-temporal subgraph signal, denoted by $\{\mathbf{x}_t[V_{\text{sub}}]\}_{t\in\mathbb{N}} = \{\mathbf{P}_{\text{sub}}\mathbf{x}_t\}_{t\in\mathbb{N}}$, where $\mathbf{P}_{\text{sub}} \in \{0, 1\}^{|V_{\text{sub}}|\times|V|}$ is the *spatial selection matrix*. By construction, $\mathbf{P}_{\text{sub}}$ satisfies $\mathbf{P}_{\text{sub}}\mathbf{P}_{\text{sub}}^\intercal = \mathbf{I}_{|V_{\text{sub}}|}$.

B. Temporal Shift of Signals

To model temporal dependencies over a finite observation window of length T, we consider linear operators acting on a spatio-temporal signal fragment $\mathbf{x}_{t_0:t_0+T} \in \mathbb{R}^{|V|\times T}$. Given the vertex-by-time structure of the signal fragment, temporal operators are naturally applied via right multiplication. Specifically, a causal and time-invariant temporal operator can be expressed as

$$\mathbf{x}_{t_0:t_0+T} \mapsto \mathbf{x}_{t_0:t_0+T}\mathbf{Q}^\intercal,$$

where the temporal shift operator (TSO) $\mathbf{Q} \in \mathbb{R}^{T\times T}$ is a lower triangular Toeplitz (LTT) matrix of the form

$$(\mathbf{Q}_T)_{ij} = \begin{cases} a_k, & i - j = k,\ 0 \le k \le T - 1, \\ 0, & \text{otherwise}, \end{cases}$$

for $i, j = 1, \ldots, T$. The lower-triangular structure encodes causality, while the Toeplitz structure reflects time invariance.

C. Algebra of Linear Operators

Given a linear operator φ, we can define its matrix polynomial algebra as

$$\mathbb{R}[\varphi] = \left\{\mathbf{M} = \sum_{i=0}^{r_0} a_i\varphi^i \, : \, a_i \in \mathbb{R}, r_0 \in \mathbb{N}\right\}.$$

If we have an operator family $\Phi = \{\varphi_1, \varphi_2, \ldots, \varphi_k\}$, we can define its free algebra $\mathbb{R}\langle\varphi_1, \varphi_2, \ldots, \varphi_k\rangle$ as

$$\operatorname{span}_{\mathbb{R}}\left\{\prod_{i=0}^{r_0}\varphi_{l_i} \, : \, \varphi_{l_i} \in \Phi, r_0 \in \mathbb{N}\right\}.$$

We use $\mathbb{R}_{\le r}[\varphi]$ or $\mathbb{R}_{\le r}\langle\varphi_1, \varphi_2, \ldots, \varphi_k\rangle$ to denote the subalgebra with at most r operators combined in a monomial.

D. Joint Domain Subgraph Filter Learning

$$\begin{tikzcd}
& (G, \mathbf{y}_{t:t+T}) \arrow[r, "\mathcal{P}_{\text{sub}}"] & (G_{\text{sub}}, \mathcal{P}_{\text{sub}}\mathbf{y}_{t:t+T}) \arrow[dd, dash, dashed, "\overset{?}{\approx}"] \\
(G, \mathbf{x}_{t:t+T}) \arrow[ur, "\mathbf{F}\otimes\mathbf{I}_T"] \arrow[dr, "\mathcal{P}_{\text{sub}}"'] & & \\
& (G_{\text{sub}}, \mathcal{P}_{\text{sub}}\mathbf{x}_{t:t+T}) \arrow[r, "\mathcal{F}_{\text{sub}}"] & (G_{\text{sub}}, \mathcal{F}_{\text{sub}}\mathcal{P}_{\text{sub}}\mathbf{x}_{t:t+T})
\end{tikzcd}$$

Fig. 1. Diagram for JST-SFL.

The JST-SFL problem is depicted in Fig. 1. Consider a spatio-temporal signal fragment $\mathbf{x}_{t:t+T} \in \mathbb{R}^{N\times T}$ defined on a graph G, where T represents the temporal window length and $T \ll t$. The stakeholder processes this signal using a spatial-only filter $\mathbf{F} \otimes \mathbf{I}_T$, followed by a subgraph extraction operation represented by $\mathcal{P}_{\text{sub}} = \mathbf{P}_{\text{sub}} \otimes \mathbf{I}_T$. A third party, with access restricted to subgraph observations, seeks to design a proprietary *spatio-temporal subgraph filter* $\mathcal{F}_{\text{sub}}$ such that the filtered output $\mathcal{F}_{\text{sub}}\mathcal{P}_{\text{sub}}\mathbf{x}_{t:t+T}$ closely approximates the target subgraph signal $\mathcal{P}_{\text{sub}}\mathbf{y}_{t:t+T}$.

Specifically, consider a training dataset comprising spatio-temporal signal fragments of the form $\mathcal{D}_{\text{train}} = \{(\mathbf{x}_{\tau:\tau+T}, \mathbf{y}_{\tau:\tau+T})\}_{\tau=0}^{t-T}$, where each fragment spans a fixed temporal window of length T. The goal of JST-SFL is to learn a subgraph-supported spatio-temporal filter $\mathcal{F}_{\text{sub}}$ that, for each training fragment, generates outputs that closely approximate the corresponding filtered subgraph signals.

This objective can be formalized as an empirical risk minimization problem:

$$\min_{\mathcal{F}_{\text{sub}}\in\mathcal{H}} \sum_{(\mathbf{x},\mathbf{y})\in\mathcal{D}_{\text{train}}} \mathcal{L}(\mathcal{F}_{\text{sub}}\mathcal{P}_{\text{sub}}\mathbf{x}, \mathcal{P}_{\text{sub}}\mathbf{y}), \tag{1}$$

where $\mathcal{H}$ represents the joint operator space of admissible vertex-time filters, which depends on T and G, and $\mathcal{L}(\cdot,\cdot)$ denotes the loss function.

To ensure scalability and mitigate the risk of over-parameterization, an explicit capacity constraint is imposed on the operator space $\mathcal{H}$. Rather than directly constraining the dimension of $\mathcal{H}$, which may not accurately reflect the number of trainable parameters, we parameterize the subgraph spatio-temporal filter family $\mathcal{H}$ as

$$\mathcal{H} = \left\{ \mathbf{H} \,:\, \mathbf{H} = \sum_{i=1}^{p} a_i \mathbf{H}_i \right\},$$

where p denotes the number of learnable parameters, and $\mathbf{H}_i$ are basis operators. The capacity constraint is then expressed as

$$p \leq \alpha N^{\beta}, \tag{2}$$

where N is the number of vertices in the ambient graph, and $\alpha > 0$ and $\beta \geq 0$ are user-defined hyperparameters that control the expressive power of the subgraph filter bank.

For instance, in an extreme case, $\mathcal{H}$ could encompass all tensors of the form $\mathbf{E}_G \otimes \mathbf{E}_T$, where $\mathbf{E}_G \in \mathbb{R}^{N\times N}$ and $\mathbf{E}_T \in \mathbb{R}^{T\times T}$ are one-hot matrices. In this scenario, the dimension of $\mathcal{H}$ would be N^2T^2. However, in practical applications, β is often set to 1 to prevent exponential growth in the parameter space.

III. Joint Domain Algebra

A. Construction of Joint Domain Algebra

Given a graph $G = (V, E)$ and its subgraph $G_{\mathrm{sub}} = (V_{\mathrm{sub}}, E_{\mathrm{sub}})$, we aim to construct a joint vertex-time domain subgraph filter bank that approximates the target filter while maintaining a controllable parameter count. To address the structural limitations of the induced subgraph, we incorporate distance-k subgraphs to capture higher-order connectivity in the ambient graph [10, Definition 1]:

Definition 1. *The distance-k subgraph of G_{sub}, denoted $(G, G_{\mathrm{sub}})^{[k]}$ or G_k, is defined as the graph with vertex set V_{sub} and edge set $\{(u, v) \in V_{\mathrm{sub}} \times V_{\mathrm{sub}} : d_G(u, v) = k\}$, where $d_G(u, v)$ is the shortest-path distance in G. Thus, two vertices are connected in G_k if they are k hops apart in G.*

The introduction of distance-k subgraphs expands the spatial filter bank associated with a given subgraph. For instance, by utilizing the symmetric normalized Laplacian $\widetilde{\mathbf{L}}_k$ of each G_k, one can construct a subgraph filter free algebra (SFFA) as $\mathscr{A}_{(G,G_{\mathrm{sub}})} = \mathbb{R}\langle \widetilde{\mathbf{L}}_1, \widetilde{\mathbf{L}}_2, \dots, \widetilde{\mathbf{L}}_{\mathrm{diam}(G)} \rangle$. These operators operate exclusively in the vertex domain and are agnostic to temporal ordering.

In contrast, temporal dependencies must be explicitly modeled. We introduce temporal propagation through causal shift operators.

Definition 2. *Given a time window T, the unit causal shift matrix $\mathbf{Q}_T \in \mathbb{R}^{T\times T}$ is defined as*

$$(\mathbf{Q}_T)_{ij} = \begin{cases} 1, & i = j+1, \\ 0, & \textit{otherwise}, \end{cases} \qquad i, j = 1, \dots, T.$$

The temporal shift polynomial algebra is then given by $\mathscr{A}_T = \mathbb{R}[\mathbf{Q}_T]$.

The central idea is to treat spatial and temporal propagation operators as algebraic generators of equal importance and to construct a joint operator space through their algebraic composition. This leads to the formulation of the joint-domain SFFA.

Definition 3. *Let G be a graph, G_{sub} its subgraph, $G_1, G_2, \dots, G_k$ the sequence of distance-k subgraphs, and T the time window. The joint-domain SFFA is defined as*

$$\begin{aligned}&\mathscr{A}_{(G,G_{\mathrm{sub}})}\\ &= \mathbb{R}\langle \widetilde{\mathbf{L}}_1 \otimes \mathbf{I}_T, \widetilde{\mathbf{L}}_2 \otimes \mathbf{I}_T, \dots, \widetilde{\mathbf{L}}_{\mathrm{diam}(G)} \otimes \mathbf{I}_T, \mathbf{I}_{V_{\mathrm{sub}}} \otimes \mathbf{Q}_T \rangle.\end{aligned}$$

Furthermore, the (k, r)-joint-SFFA is defined as

$$\begin{aligned}&\mathscr{A}_{(G,G_{\mathrm{sub}})}(k, r)\\ &= \mathbb{R}_{\leq r}\langle \widetilde{\mathbf{L}}_1 \otimes \mathbf{I}_T, \widetilde{\mathbf{L}}_2 \otimes \mathbf{I}_T, \dots, \widetilde{\mathbf{L}}_k \otimes \mathbf{I}_T, \mathbf{I}_{V_{\mathrm{sub}}} \otimes \mathbf{Q}_T \rangle.\end{aligned}$$

By replacing $\widetilde{\mathbf{L}}_k$ with $\widetilde{\mathbf{L}}_k^{1/k}$, a fractional-powered version of the joint SFFA can be obtained.

This construction extends the SFFA framework developed for standard SFL by incorporating the joint vertex-time domain and introducing a temporal shift operator. The joint-domain SFFA enables the modeling of time-dependent information propagation, which cannot be captured by the original SFFA, regardless of its spatial complexity. Moreover, the joint-domain SFFA remains scalable, as its size can be controlled by adjusting the parameters k and r, consistent with the principles of linear operator algebra.

B. Parameter Analysis

We now analyze the number of trainable parameters in the (k, r)-joint-SFFA.

Theorem 1. *For a graph G and its subgraph G_{sub}, given a temporal window of length T, the number of trainable parameters in the (k, r)-joint-SFFA is given by $\frac{1}{k}((k+1)^{r+1} - 1)$.*

Proof. The (k, r)-joint-SFFA is generated by $k + 1$ linear operators: k Laplacian-based spatial operators and one temporal shift operator. For any truncation order r, the total number of distinct operator combinations, including repetitions, is $\sum_{l=0}^{r} (k + 1)^l$, where l denotes the number of operators in a monomial. Each combination corresponds to a trainable parameter. Evaluating the geometric sum yields

$$\sum_{l=0}^{r} (k+1)^l = \frac{(k+1)^{r+1} - 1}{(k+1) - 1} = \frac{1}{k}((k+1)^{r+1} - 1).$$

This completes the proof. $\square$

The flexibility of the (k, r)-joint-SFFA lies in its ability to balance spatial and temporal expressiveness within a constrained parameter budget. Notably, the number of trainable parameters in the (k, r)-joint-SFFA matches that of a purely spatial $(k + 1, r)$-SFFA, highlighting an explicit trade-off between spatial operator diversity and temporal modeling. Furthermore, since the parameters k and r are independent of both the graph size and the temporal window length T, the (k, r)-joint-SFFA provides a scalable and adaptable framework for constructing filter banks. This scalability ensures that the framework remains computationally efficient while offering sufficient flexibility to model complex spatio-temporal dependencies.

IV. Experiments

A. JST-SFL of Temporal-Recursive Graph Signal on Grid Networks

To evaluate the performance of the proposed joint SFFA framework in the context of JST-SFL, we conduct experiments on a 10×10 grid graph G. The graph G consists of $N = 100$ nodes, with vertex set V, a subset V_{sub} (where $|V_{\text{sub}}| = pN$, and p denotes the subgraph ratio), a selection matrix $\mathbf{P}_{\text{sub}} \in \{0,1\}^{|V_{\text{sub}}| \times N}$, the normalized ambient graph Laplacian $\widetilde{\mathbf{L}} \in \mathbb{R}^{N \times N}$, and the normalized subgraph Laplacian $\widetilde{\mathbf{L}}_0 \in \mathbb{R}^{|V_{\text{sub}}| \times |V_{\text{sub}}|}$.

We generate a polynomial graph filter $\mathbf{F} = 0.5\mathbf{I} - \widetilde{\mathbf{L}} + 2\widetilde{\mathbf{L}}^2 + \widetilde{\mathbf{L}}^3$. For each trial, a random subset of p proportion of nodes is selected to form the induced subgraph G_{sub}.

The spatio-temporal graph signal sequence is generated as follows: at $t = 0$, each node's signal is independently sampled from $\text{Unif}\,(-5, -3)$, and at $t = 1$, from $\text{Unif}\,(2, 7)$. Subsequent signals are recursively generated using the relation $\mathbf{x}_t = \mathbf{x}_{t-1} - \mathbf{x}_{t-2} + 0.2\mathcal{N}(0,1)$, producing a sequence of length $t = 130$. The first 100 time steps are used for training, while the remaining 30 time steps are reserved for testing. The objective is to train a joint filter $\mathcal{F}_{\text{sub}}$ within a specified filter space using the training data, such that the average recovery error on V_{sub} (as defined in (1)) is minimized. The maximum number of trainable parameters is constrained to $1.0 \times N^1 = 100$.

Performance is evaluated using the mean squared error (MSE) on the test sequence. Spatial filters are applied to individual snapshots, while spatio-temporal filters are applied using a causal sliding window of length $T = 6$ over the sequence. The test error corresponding to (1) is defined as:

$$e_{\text{test}} := \frac{1}{30 \cdot |V_{\text{sub}}|} \sum_{t=101}^{130} \big\| \mathbf{P}_{\text{sub}} \mathbf{F} \mathbf{x}_t - \big(\mathcal{F}_{\text{sub}} (\mathbf{P}_{\text{sub}} \otimes \mathbf{I}_T) \mathbf{x}_{t-T+1:t+1} \big)_t \big\|_2^2. \tag{3}$$

We compare the performance of the proposed method against several baseline filter spaces, including:

- Truncated filter $\mathbf{F}_{\text{trunc}} = \mathbf{P}_{\text{sub}} \mathbf{F} \mathbf{P}_{\text{sub}}^{\intercal}$, a non-trainable baseline.
- Subgraph Laplacian polynomial filter space $\mathbb{R}_{\leq 3}[\widetilde{\mathbf{L}}_0]$, with 4 trainable parameters.
- $(3,3)$-SFFA, a purely spatial filter space with 40 trainable parameters.
- $(4,3)$-SFFA, a purely spatial filter space with 85 trainable parameters.
- $(3,3)$-joint-SFFA, the proposed joint-domain filter space, with 85 trainable parameters.

For all filter spaces except the truncated filter, the coefficients are learned during training. The Adam optimizer is employed with a learning rate of 0.01 for 2000 epochs to ensure convergence. Each experiment is repeated five times, and the average test MSE is reported.

TABLE I
Test MSE on the synthetic grid dataset for different subgraph sizes.

Method	$p = 0.2$	$p = 0.3$	$p = 0.4$
Truncated Filter	439.03	384.38	325.29
Laplacian Polynomial	109.72	82.48	74.66
$(3,3)$-SFFA	63.54	53.12	42.53
$(4,3)$-SFFA	**42.78**	38.92	33.42
$(3,3)$-joint-SFFA	48.63	**38.46**	**28.40**

The experimental results presented in Table I demonstrate the effectiveness of the $(3,3)$-joint-SFFA framework. Specifically, the following observations can be made:

- The $(3,3)$-joint-SFFA achieves performance comparable to or exceeding that of the $(4,3)$-SFFA, despite operating under an equivalent parameter budget.
- The inclusion of temporal modeling in the $(3,3)$-joint-SFFA leads to superior performance compared to the $(3,3)$-SFFA, which utilizes the same set of distance-k subgraphs and maximum polynomial degree but lacks temporal domain integration.
- The $(3,3)$-joint-SFFA consistently outperforms other spatial-only baselines, including the induced subgraph Laplacian polynomial filter and the non-trainable truncated filter.

B. JST-SFL of Graph Signal on METR-LA Dataset

TABLE II
Test MSE on the METR-LA traffic dataset for different subgraph sizes.

Method	$p = 0.2$	$p = 0.3$	$p = 0.4$
Truncated Filter	16863.05	13028.08	11891.61
Laplacian Polynomial	549.22	547.37	552.77
$(3,3)$-SFFA	254.27	292.97	307.71
$(4,3)$-SFFA	160.23	237.71	279.03
$(3,3)$-joint-SFFA	**157.28**	**216.23**	**235.52**

The experimental setup on the METR-LA traffic dataset [11] mirrors the configuration used in the synthetic grid experiments, including identical filter specifications, parameter constraints, and training protocols. Note that it is an operator learning task rather than spatio-temporal forecasting. The results, summarized in Table II, demonstrate that the proposed $(3,3)$-joint-SFFA consistently achieves superior performance compared to all other subgraph filter methods, highlighting its effectiveness in capturing spatio-temporal dependencies.

V. Conclusion

This paper introduced a joint vertex–time formulation for subgraph filter learning, leveraging a structured operator algebra with explicit capacity constraints. By incorporating causal temporal shift operators alongside spatial subgraph operators as algebraic generators, the proposed framework facilitates efficient and interpretable spatio-temporal filtering on subgraphs. Experimental evaluations on both synthetic datasets and real-world traffic data demonstrate that the joint-domain approach consistently outperforms spatial-only subgraph filter learning methods, achieving superior performance within comparable parameter constraints.

REFERENCES

[1] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE Trans. Neural Netw.*, vol. 20, no. 1, pp. 61–80, Jan. 2009.

[2] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *International Conference on Learning Representations*, 2017.

[3] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Advances in Neural Inform. Process. Syst.*, USA, 2016, pp. 3844–3852.

[4] X. Dong, D. Thanou, P. Frossard, and P. Vandergheynst, "Learning Laplacian matrix in smooth graph signal representations," *IEEE Trans. Signal Process.*, vol. 64, no. 23, pp. 6160–6173, Dec. 2016.

[5] H. E. Egilmez, E. Pavez, and A. Ortega, "Graph learning from data under Laplacian and structural constraints," *IEEE J. Sel. Top. Signal Process.*, vol. 11, no. 6, pp. 825–841, Sep. 2017.

[6] S. Chen, A. Sandryhaila, J. M. F. Moura, and J. Kovacevic, "Signal denoising on graphs via graph filtering," in *Proc. IEEE Global Conf. on Signal and Information Processing*, Dec. 2014, pp. 872–876.

[7] F. Ji, W. P. Tay, and A. Ortega, "Graph signal processing over a probability space of shift operators," *IEEE Trans. Signal Process.*, vol. 71, pp. 1159–1174, 2023.

[8] S. Chen, A. Sandryhaila, J. M. F. Moura, and J. Kovačević, "Signal recovery on graphs: Variation minimization," *IEEE Trans. Signal Process.*, vol. 63, no. 17, pp. 4609–4624, Sep. 2015.

[9] A. Sandryhaila and J. M. F. Moura, "Discrete signal processing on graphs: Graph fourier transform," in *Proc. IEEE Int. Conf. Acoustics, Speech, and Signal Processing*, May 2013, pp. 6167–6170.

[10] P. Zhang, F. Ji, Y. Zhao, W. P. Tay, and B. Wen, "Distance-based laplacian algebra for effective subgraph filter learning," in *2025 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, Oct 2025, pp. 1301–1306.

[11] Y. Li, R. Yu, C. Shahabi, and Y. Liu, "Diffusion convolutional recurrent neural network: Data-driven traffic forecasting," in *International Conference on Learning Representations*, 2018.